

# Xuanyu Zhou

🏠 3600 Chestnut Street Philadelphia, PA 19104

✉ xuanyuz@design.upenn.edu 📞 (510) 570-5958 💻 xuanyuzhou.org 🌐 GitHub

**Research interests:** urban analytics, built environment, urban mobility, data science, Large Language Models

## Education

### University of Pennsylvania

*Master of City Planning*

**Philadelphia, PA**

2026–2028 (Exp.)

### Zhejiang University

*B.E. in Urban and Rural Planning, 5 year program 3.93/4.0*

**Hangzhou, Zhejiang**

2021–2026

**Coursework:** Principle of Urban Planning, Urban Transportation, Architecture Design, Micro/Macroeconomics

### University of California, Berkeley

*Exchange in College of Environmental Design*

**Berkeley, CA**

2025

**Coursework:** Introduction to City Planning, The Structure and Interpretation of Computer Programs, Transit

## Research Experiences

### MIT-UF-NEU 2025 Joint Summer Research Camp

**Remote**

*Mentee, mentor: Dr. Xuan Jiang*

Jun 2025–Oct 2025

- Conducted comprehensive result analysis for a novel GPU-based computational framework for large-scale Urban Air Mobility (UAM) demand modeling and heterogeneous fleet optimization, applied to a case study in the Bay Area.
- Produced spatial and temporal demand visualizations, including geographic heatmaps and fleet-type comparisons, using Python, ArcGIS, Matplotlib, and Plotly.
- Authored the full first draft of the research manuscript as second author, synthesizing findings that demonstrated over 20 minutes of travel time savings for 230,000 selected trips.

### Monitoring-Mapping-Modeling (3M) Lab, UC Berkeley

**Berkeley, CA**

*Research Assistant, supervisor: Prof. Lu Liang*

Mar 2025–Present

- Designed and implemented a Python-based pipeline to compute urban Sky View Factor (SVF) from street view imagery (SVI) by converting 2D images to 3D point clouds via MoGe, enabling SVF estimation at pedestrian-level positions on sidewalks rather than road-center viewpoints used in conventional SVI approaches.
- Processed over 120,000 street view images across four U.S. cities (San Francisco, Los Angeles, Dallas, Seattle), with planned expansion to major cities nationwide. Conducted cross-source validation comparing SVF derived from SVI, LiDAR, and image-based point clouds.
- Research addresses thermal equity by providing a scalable alternative to coverage-limited LiDAR data, extending accurate SVF and heat exposure estimation to underserved regions including the Global South.

### Urban Greenery Equity Research Project

**Hangzhou, Zhejiang**

*Independent Research, advised by Mingze Chen*

Jun 2024–Oct 2025

- Independently conducted a longitudinal study on urban green space (UGS) equity in Pittsburgh, a severely shrinking city, analyzing satellite imagery from 2019 to 2022 using machine learning methods to assess changes in UGS distribution, area, and morphology.
- Processed census-tract-level visitation data and applied OLS and Geographically Weighted Regression (GWR) to examine how UGS characteristics influence community vitality in shrinking neighborhoods.
- Authored the full first draft of the manuscript as first author. Presented findings as an oral presentation at the 2024 ACSP Annual Conference.

### Qizhenwenxue Undergraduate Research Program, Zhejiang University

**Hangzhou, Zhejiang**

*Research Assistant, supervisor: Prof. Shuang Ma*

Aug 2023–Nov 2025

- Authored the Introduction and Literature Review, establishing the theoretical framework on urban mobility and resilience in Nagoya. Contributed to data preprocessing and Geographic Random Forest (GRF) modeling using mobile phone signaling data. Published in *Transactions in Urban Data, Science, and Technology*.

- Independently implemented GRF modeling, including parameter tuning and feature importance analysis, on a dataset of over 600,000 geotagged Weibo posts across four months. Interpreted resulting spatial patterns linking built environment features to resident mental health indicators. Manuscript under review at *Urban Studies*.
- Conducted GRF modeling pipeline and data visualization on over 9 million shared bicycle trip records in Shenzhen. Contributed to result interpretation and manuscript conclusion. Under review at *npj Urban Sustainability*.

## Publications

### Published

[5] Ma, S., **Zhou, X.**, Cai, W., Chen, M., Li, S. (2025). Evaluating the social-economic recovery impacts of built environment for post-pandemic. *Transactions in Urban Data, Science, and Technology*.

### Under Review

[4] Chen, M., **Zhou, X.**, & Liu, Y. How do we model cities? A systematic review of spatial and nonlinear analytical paradigms in planning research. *Landscape and Urban Planning*.

[3] Jiang, X., **Zhou, X.**, Zhao, Y., Cao, S., He, H., Zhao, J., Hansen, M., Sengupta, R. From Patchwork to Network: A Comprehensive Framework for Demand Analysis and Fleet Optimization of Advanced Air Mobility. *IEEE Transactions on Intelligent Transportation Systems*.

[2] Ma, S., Li, W., **Zhou, X.**, Ma, T., He, Y., & Li, S. Future life expectancy extension under bike ride, air pollution exposure, and traffic accidents through modifying the built environment and climate: An integrated modelling study. *npj Urban Sustainability*.

[1] Li, S., Cai, W., Wu, C., Zhang, X., **Zhou, X.**, He, Y., & Ma, S. Exploring the relationship between resident emotions and built environment in Beijing based on a geographical weighted random forest approach. *Urban Studies*.

### Conference Presentations

[C3] Zhou, Y., **Zhou, X.**, & Liang, L. (2026). Perceiving the Street from Where People Stand: Pedestrian-Centered Multi-View-Factor Mapping via Street View Imagery and Monocular 3D Reconstruction. *33rd International Conference on Geoinformatics, Singapore*. (Oral Presentation)

[C2] Jiang, X., **Zhou, X.**, Zhao, Y., Cao, S., He, H., Zhao, J., Hansen, M., Sengupta, R. (2026). From Patchwork to Network: A Comprehensive Framework for Demand Analysis and Fleet Optimization of Urban Air Mobility. *2026 Transportation Research Board Annual Meeting*. (Oral Presentation)

[C1] **Zhou, X.**, & Chen, M. (2024). Greening the Gap: Examining Urban Greenery Equity in Shrinking Cities. *2024 ACSP Annual Conference*. (Oral Presentation)

## Internships

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| <b>Zhejiang University Urban—Rural Planning &amp; Design Institute Co., Ltd.</b>                                                                                                                                                                                                                                                                                                                                                                                       | <b>Hangzhou, Zhejiang</b> |
| <i>Assistant Planner</i>                                                                                                                                                                                                                                                                                                                                                                                                                                               | Aug 2025–Nov 2025         |
| <ul style="list-style-type: none"> <li>- Conducted full life-cycle carbon emission assessment for the Anji Economic Development Zone’s national-level Zero-Carbon Industrial Park application, following Chinese national standards.</li> <li>- Performed GIS-based site selection analysis for low-altitude economy infrastructure across three municipal planning projects; produced spatial visualizations and contributed to stakeholder consultations.</li> </ul> |                           |

## Skills

|                                    |                                                                                            |
|------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Programming &amp; Data:</b>     | Python (pandas, NumPy, geopandas, scikit-learn, Matplotlib, Plotly), R, SQL, Git, $\LaTeX$ |
| <b>Geospatial &amp; Analysis:</b>  | ArcGIS, QGIS, Google Earth Engine, Machine Learning                                        |
| <b>Design &amp; Visualization:</b> | Rhino, SketchUp, Adobe Suite (Photoshop, Illustrator, InDesign)                            |

## Awards

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|----------------------------------------------------------------------------------------------|------------------|
| <b>Zhejiang Provincial Government Scholarship</b> , Zhejiang Province (top 3%)               | 2024, 2025       |
| <b>Outstanding Graduate</b> , Zhejiang University                                            | 2026             |
| <b>Outstanding Undergraduate Thesis</b> , Zhejiang University                                | 2026             |
| <b>Second-Prize Scholarship</b> , Zhejiang University (top 10%)                              | 2023, 2024, 2025 |
| <b>Ningbo Future Star Scholarship</b> , Human Resources and Social Security Bureau of Ningbo | 2025             |